

Kubernetes Deployment Rollout Stuck Runbook

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Category	kubernetes	Severity	P2 - High
Domain	Container Orchestration	Est. Duration	15-30 minutes
Tags	kubernetes, deployment, rollout, stuck, pending, imagepullbackoff		

Overview

A stuck rollout means a Deployment has started updating pods but the new ReplicaSet cannot reach its desired replica count. Old pods remain Running while new pods are stuck in Pending, ImagePullBackOff, or CrashLoopBackOff. This blocks all future deployments until resolved.

Prerequisites

- kubectl configured for the target cluster
- Access to container registry credentials
- Deployment history access (kubectl rollout history)

Response Steps

Step 1: Check Rollout Status

Confirm the rollout is stuck and identify which deployment version is affected.

```
kubectl rollout status deployment/ -n
kubectl rollout history deployment/ -n
```

NOTE: A stuck rollout shows 'Waiting for deployment to finish' indefinitely.

Step 2: Inspect New ReplicaSet Pods

Find the new ReplicaSet created by the rollout and check why its pods are not starting.

```
kubectl get rs -n | grep
kubectl get pods -n -l app=
kubectl describe pod -n
```

Step 3: Check for ImagePullBackOff

If the new pod shows ImagePullBackOff, the image tag does not exist in the registry or credentials are missing.

```
kubectl get pod -n -o jsonpath='{.status.containerStatuses[0].image}'
kubectl get secret regcred -n
kubectl describe pod -n | grep -A5 'Events'
```

NOTE: Fix: correct the image tag or re-create the imagePullSecret.

Step 4: Check for Resource Quota Exhaustion

If pods are stuck in Pending, the namespace may have hit its ResourceQuota for CPU or memory.

```
kubectl describe resourcequota -n
kubectl get events -n | grep FailedCreate
kubectl describe nodes | grep -A5 'Allocated resources'
```

Step 5: Check PodDisruptionBudget

A PDB with minAvailable = replicas will block the rollout from terminating old pods.

```
kubectl get pdb -n
kubectl describe pdb -n
```

NOTE: If PDB is blocking, temporarily patch minAvailable to allow the rollout to proceed.

Step 6: Pause and Resume or Rollback

If the root cause is fixed, resume the rollout. If not fixable quickly, roll back to the previous version.

```
# To resume after fixing image/quota:
kubectl rollout resume deployment/ -n

# To roll back:
kubectl rollout undo deployment/ -n
```

Rollback Steps

Rollback Step 1: Roll Back to Previous Version

Roll back to the last known good revision.

```
kubectl rollout undo deployment/ -n
kubectl rollout status deployment/ -n
```

Rollback Step 2: Pin to Specific Revision

If the immediate previous version also had issues, roll back to a specific revision.

```
kubectl rollout undo deployment/ -n --to-revision=
kubectl rollout history deployment/ -n
```

Step Dependencies

Step	Depends On	Can Parallelize With
Step 1: Check Rollout Status	—	—
Step 2: Inspect New RS Pods	Step 1	—
Step 3: Check ImagePullBackOff	Step 2	Step 4
Step 4: Check Resource Quota	Step 2	Step 3
Step 5: Check PDB	Step 2	Step 3, Step 4
Step 6: Pause/Resume/Rollback	Step 3, Step 4, Step 5	—

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